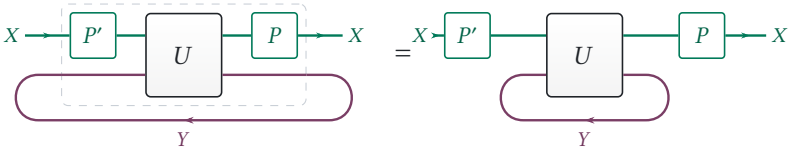
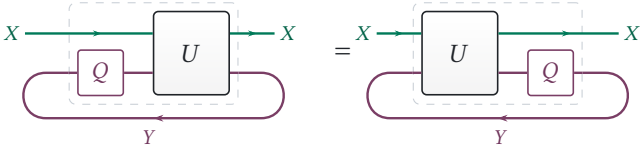
(a) naturality



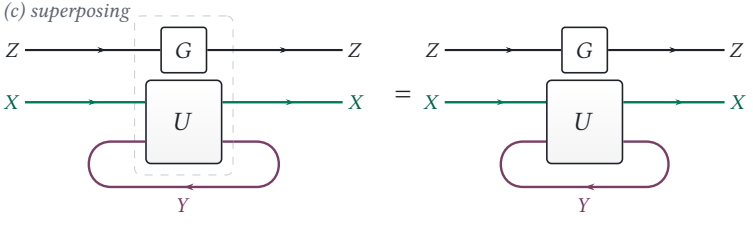
$$\mathrm{Tr}^{Y}\big((P \oplus I)\,U\,(P' \oplus I)\big) = P\ \mathrm{Tr}^{Y}(U)\,P'$$

(b) sliding



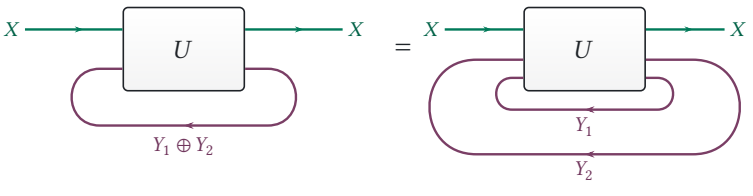
$$\mathrm{Tr}^{Y}\big(U\,(I \oplus Q)\big) = \mathrm{Tr}^{Y}\big((I \oplus Q)\,U\big)$$

(c) superposing



$\mathrm{Tr}^{Y}(G \oplus U) = G \oplus \mathrm{Tr}^{Y}(U)$, the loop taken over Y

(d) vanishing



$$\mathrm{Tr}^{Y_1 \oplus Y_2}(U) = \mathrm{Tr}^{Y_2}\big(\mathrm{Tr}^{Y_1}(U)\big)$$